Foundations of Large-Scale Multimedia Information Management and Retrieval

Lecture #1 Introduction

Edward Y. Chang

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Foundations of Large-Scale Multimedia Information Management and Retrieval
Mathematics of Perception

Foundations of Large-Scale Multimedia Information Management and Retrieval *Mathematics of Perception* covers knowledge representation and semantic analysis of multimedia data and scalability in signal extraction, data mining, and indexing. The book is divided into two parts: Part I - Knowledge Representation and Semantic Analysis focuses on the key components of mathematics of perception as it applies to data management and retrieval. These include feature selection/reduction, knowledge representation, semantic analysis, distance function formulation for measuring similarity, and multimodal fusion. Part II - Scalability Issues presents indexing and distributed methods for scaling up these components for high-dimensional data and Web-scale datasets. The book presents some real-world applications and remarks on future research and development directions.

The book is designed for researchers, graduate students, and practitioners in the fields of Computer Vision, Machine Learning, Large-scale Data Mining, Database, and Multimedia Information Retrieval.

Dr. Edward Y. Chang was a professor at the Department of Electrical & Computer Engineering, University of California at Santa Barbara, before he joined Google as a research director in 2006. Dr. Edward Y. Chang received his M.S. degree in Computer Science and Ph.D degree in Electrical Engineering, both from Stanford University.

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Mathematics of Perception

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Datasets



- YouTube
- Facebook Photos
- Netflix Movies
- Apple Music
- Archive.org
  - http://www.archive.org/details/audio
  - 400k recordings







Technical Challenges

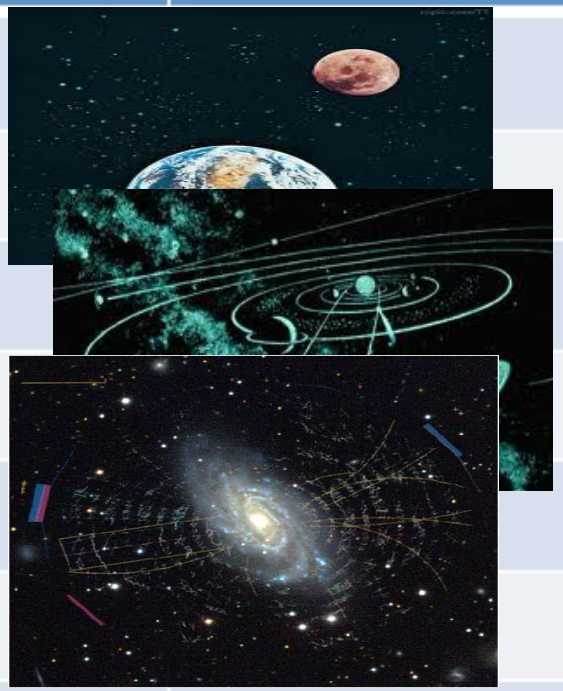
- Volume, both too small and too large
- Variety, text, video, image, music, social, etc.
- Velocity

Prefixes

SI prefix	Name	Power of 10 or 2	
k kilo	thousand	10^3 2^{10}	
M mega	million	10^6 2^{20}	
G giga	billion	10^9 2^{30}	
T tera	trillion	10^{12} 2^{40}	
P peta	quadrillion	10^{15} 2^{50}	
E exa	quintillion	10^{18} 2^{60}	
Z zetta	sextillion	10^{21} 2^{70}	
Y yotta	septillion	10^{24} 2^{80}	

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Key Applications

Use Image as an Example

- Content-based Retrieval
  - Given an image, find perceptually similar ones
- Annotation
  - Given an image, identify what (objects, events), where (location), when (time), and who (people)
- Do the above both quickly and accurately

- Applicable to any multimedia data types

Key Subroutines

- Feature Extraction
- Machine Learning
  - Small sample pool
  - Large sample pool
- Similarity
- Multimodal Fusion
- Indexing
- Scalability
  - Parallel algorithms
  - Online algorithms

Feature Extraction

- Representative Methods
  - Color, texture, shape
  - SIFT
  - Bio-motivated HMAX
  - Deep Learning
- Model-Based vs. Data-Driven

Classification

- Query-Concept Learning
  - Online
  - Concept-dependent learning
  - Imbalanced data
  - Large D
  - D >> N
  - $N^- >> N^+$
- Annotation
  - Offline
  - Feature-to-semantics mapping
  - Large N

Similarity

- Traditional Distance Function
- Dynamic Partial Function
- Learning Similarity from Data

Multimodal Fusion

- Feature Fusion (Cross-Modality)
  - Weighted sum
  - PCA
  - Dimensionality curse
- Personalization (Cross-Domain)
  - Latent behavior
- Context + Content
  - Location, time, et al.

Storage and Indexing

- Dimensionality Curse
- High-Dimensional Indexing

Large-Scale Learning

3V Problem

- Huge Volume of Data
- High Velocity
- High Variety

Key Parallel Algorithmic Work

- Parallel Spectral Clustering, W.-Y. Chen, Y. Song, H. Bai, Chih-Jen Lin, and E. Y. Chang, IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI), 2010.
- PLDA+: Parallel Latent Dirichlet Allocation with Data Placement and Pipeline Processing, ACM Transactions on Intelligent Systems and Technology, 2011.
- Parallel Support Vector Machines, E. Y. Chang, et al., NIPS 2007.
- PFP: Parallel FP-Growth for Query Recommendation, H. Li, Y. Wang, D. Zhang, M. Zhang, and E. Y. Chang, ACM Recommendation Systems, Lausanne, October 2008

More Data vs. Better Algorithms

Banko & Brill, 2001

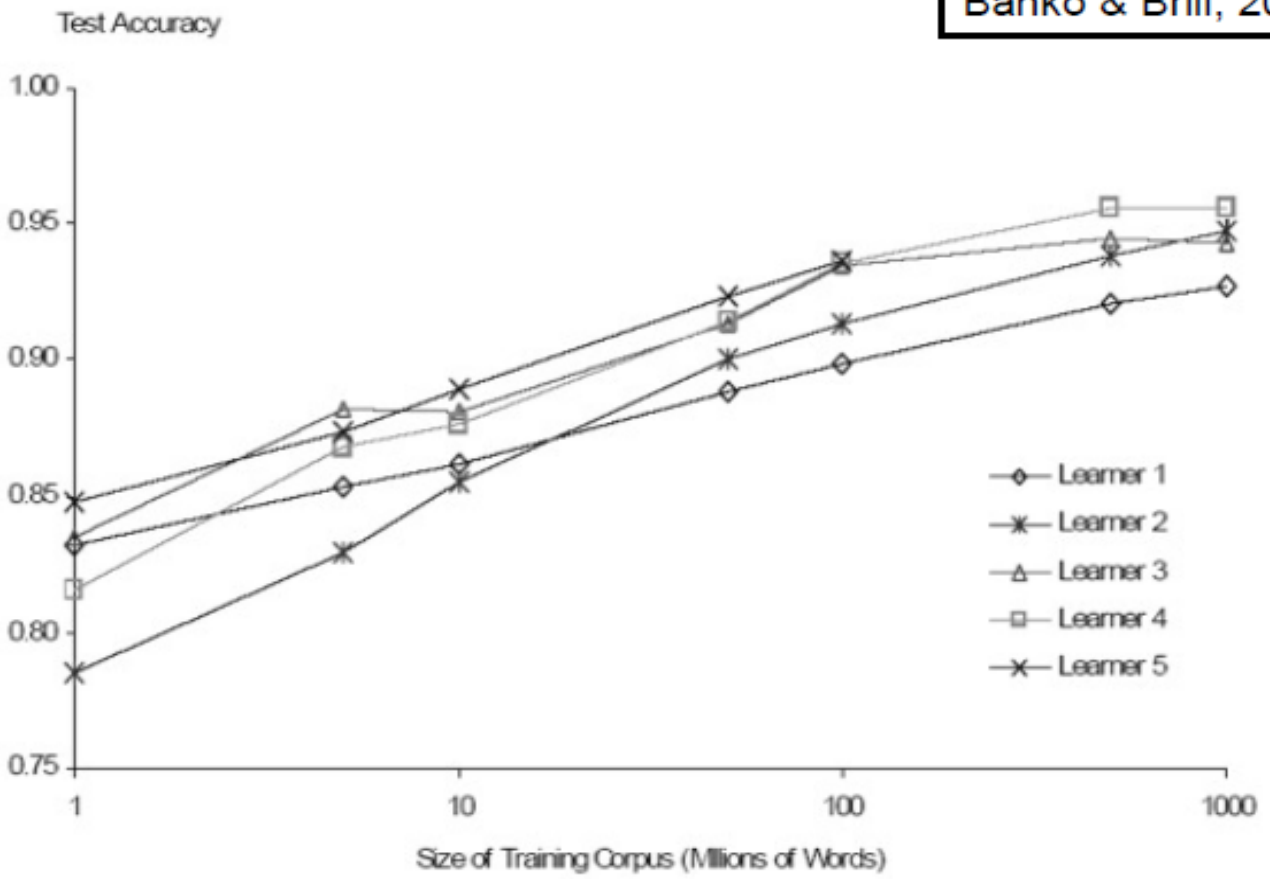


Figure 2. Learning Curves for Confusable Disambiguation

More Data vs. Better Algorithms

Banko & Brill, 2001

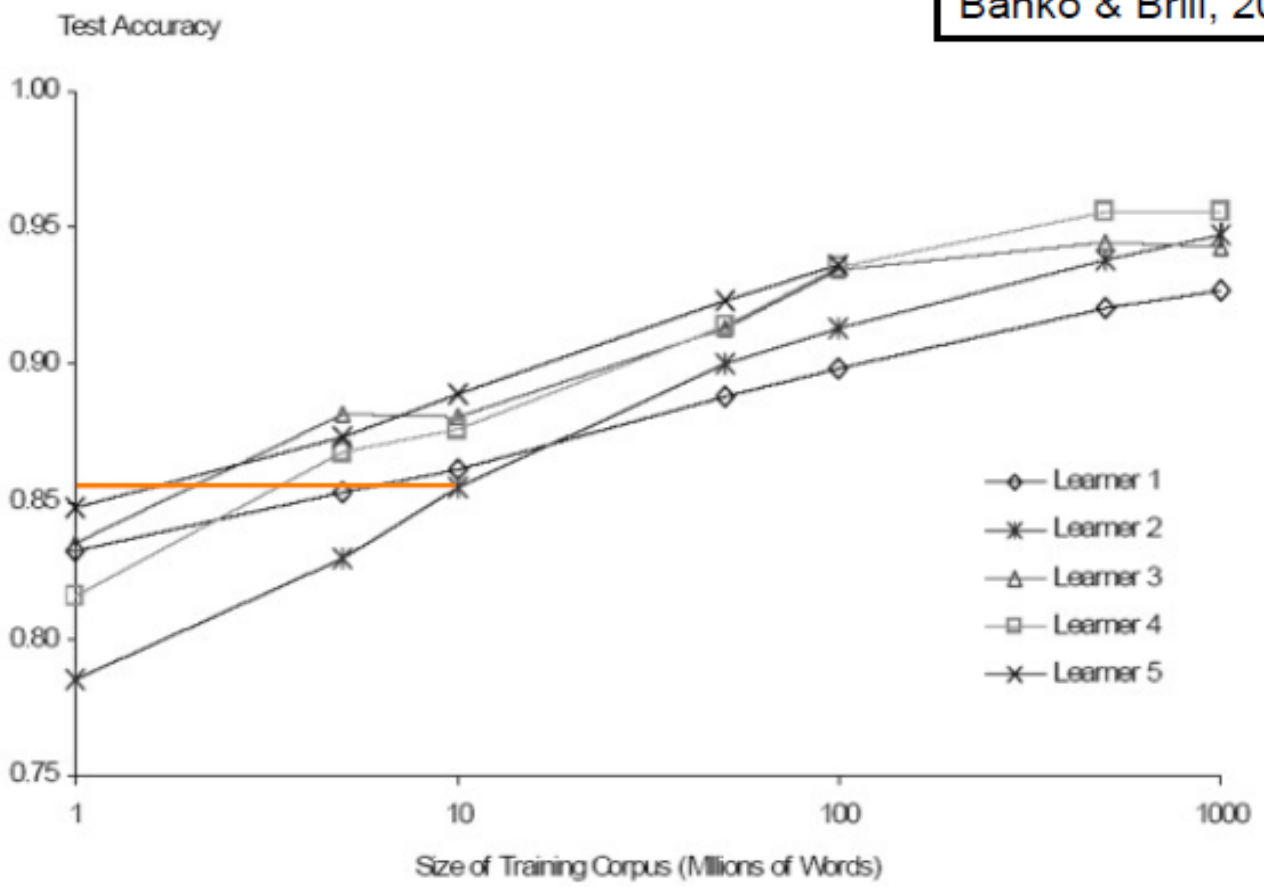


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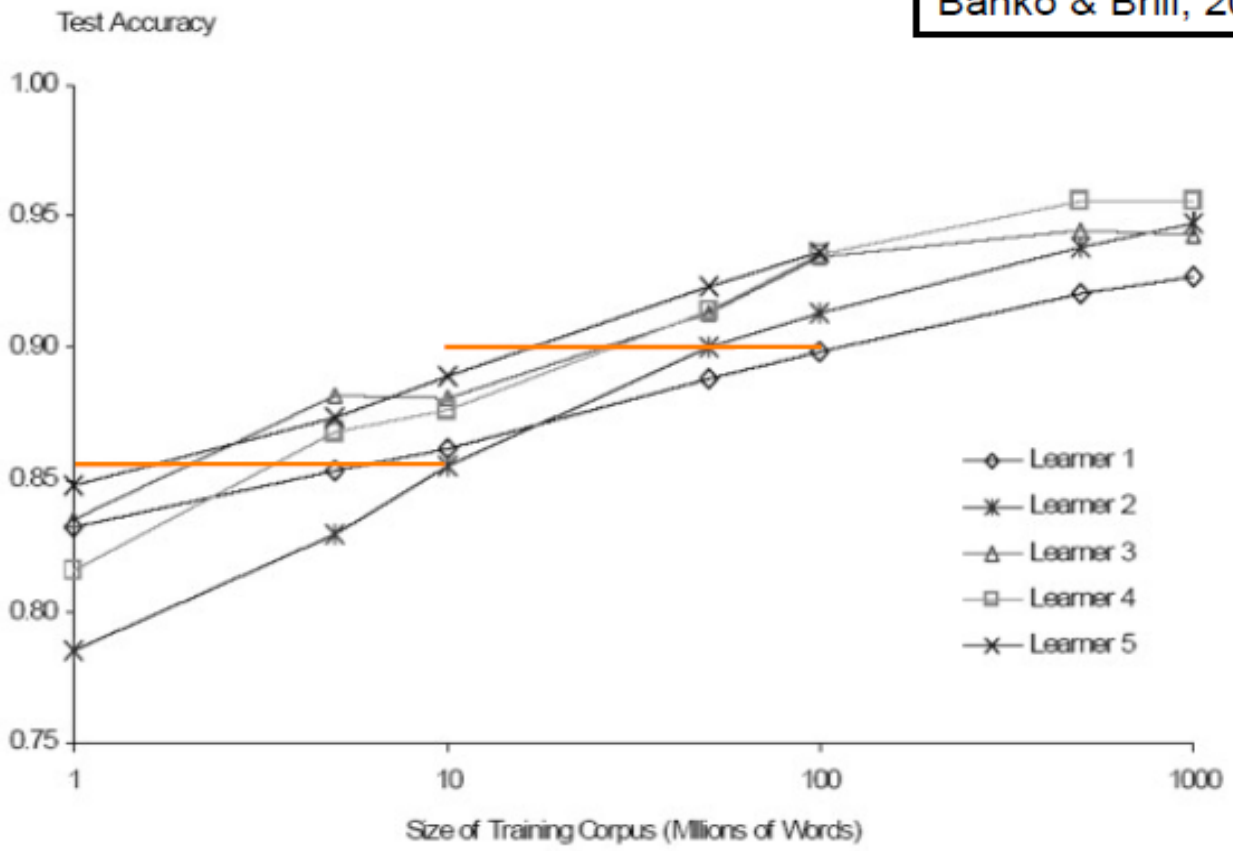


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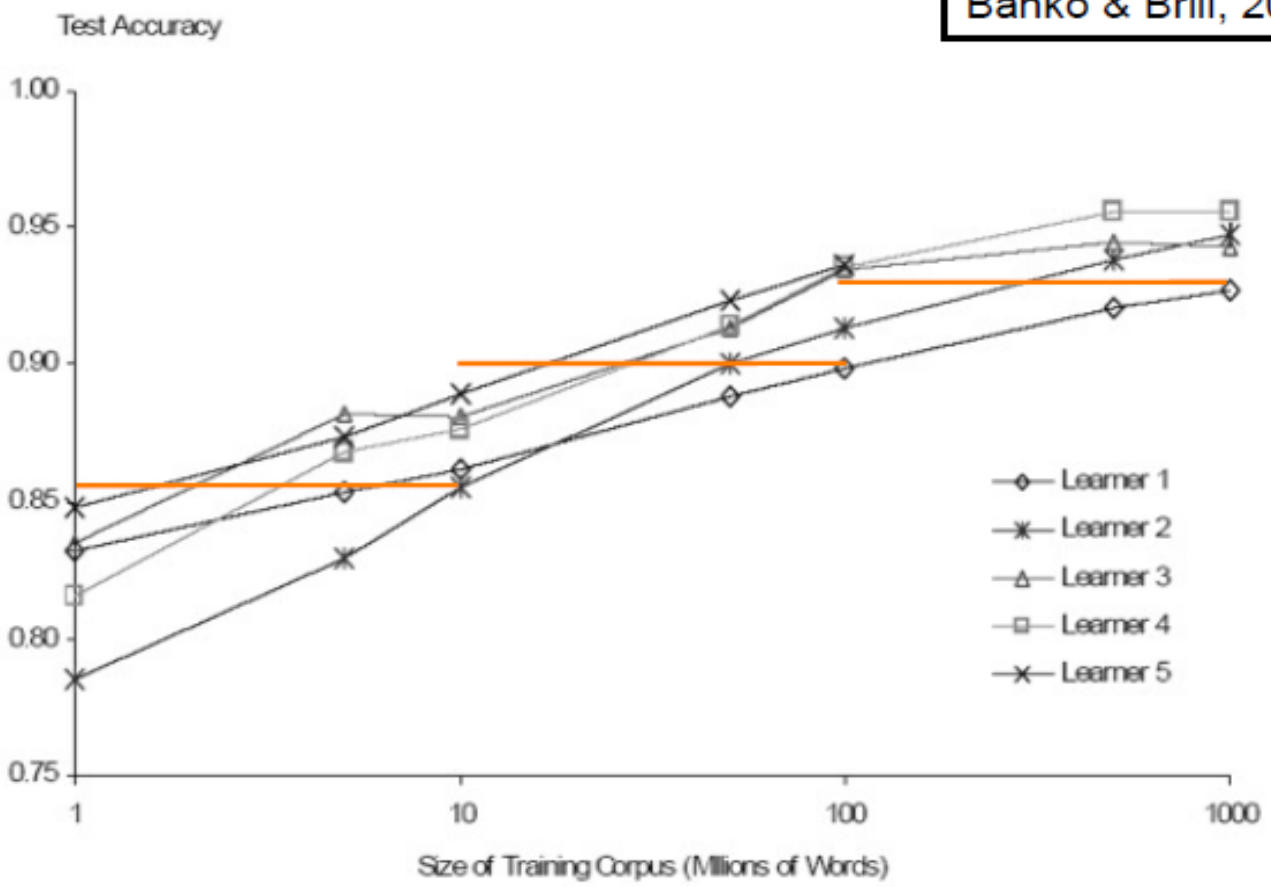


Figure 2. Learning Curves for Confusable Disambiguation

Comparative Study on Small Dataset

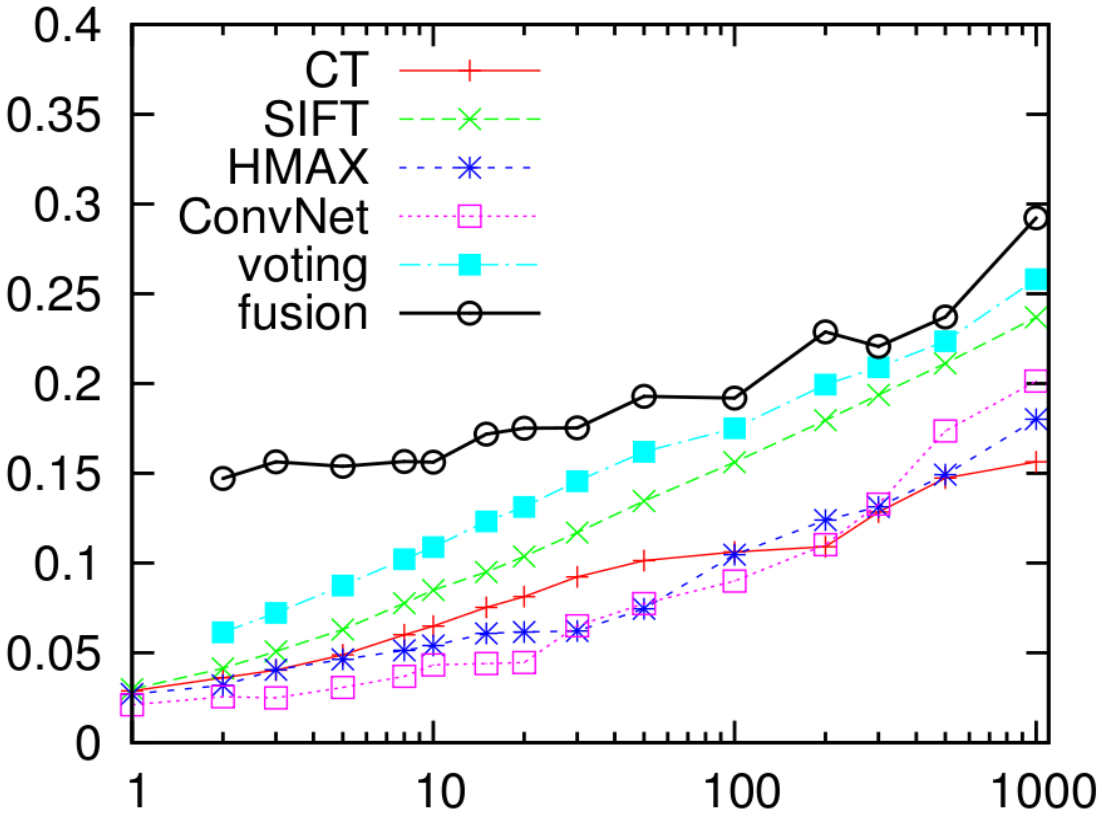
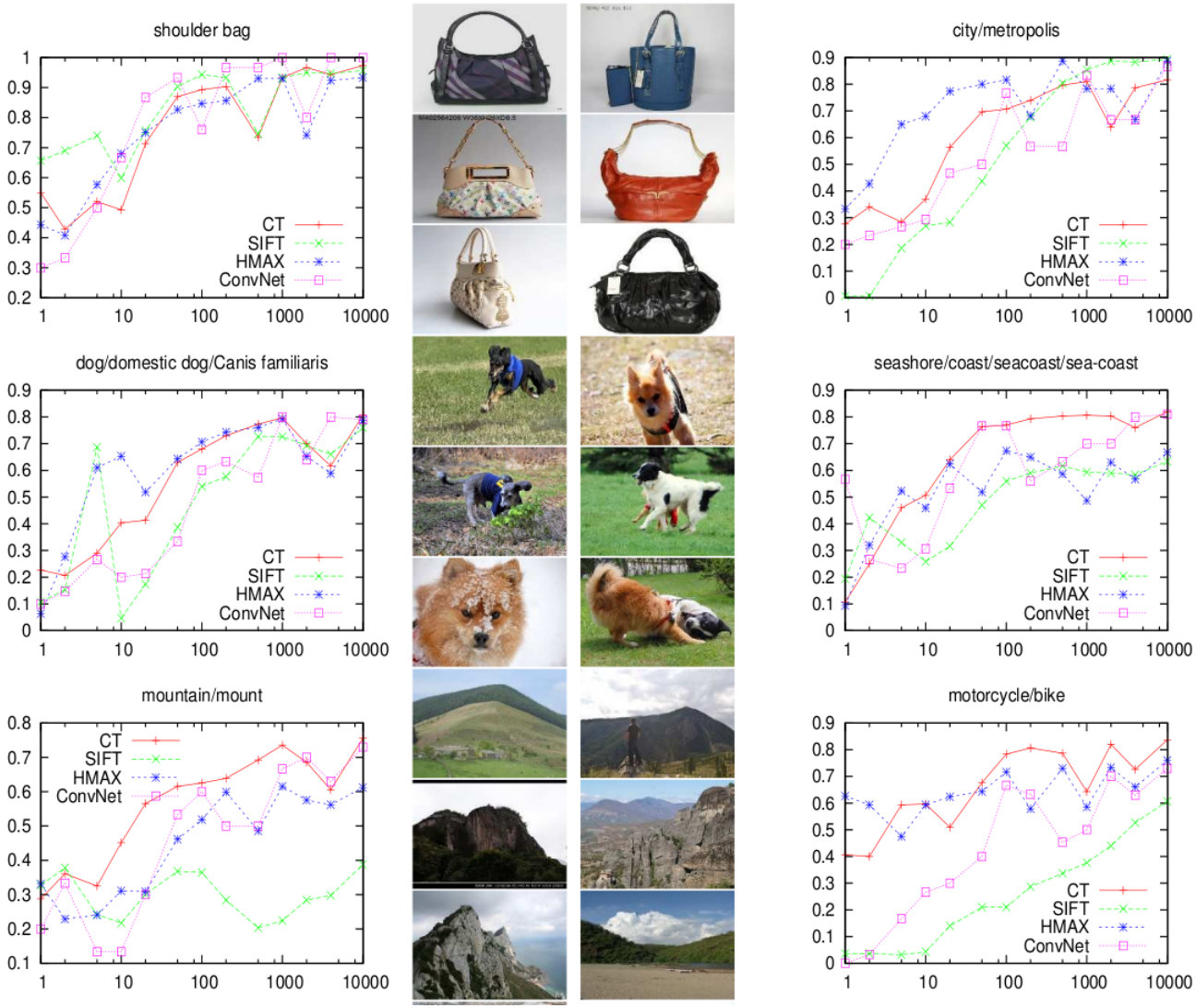


Fig. 7: Accuracy for ImageNet dataset. X-axis shows the number of training instances for each category. Y-axis shows the average accuracy.

Large Dataset

10,000 instances per class



Comparative Study on Large Dataset

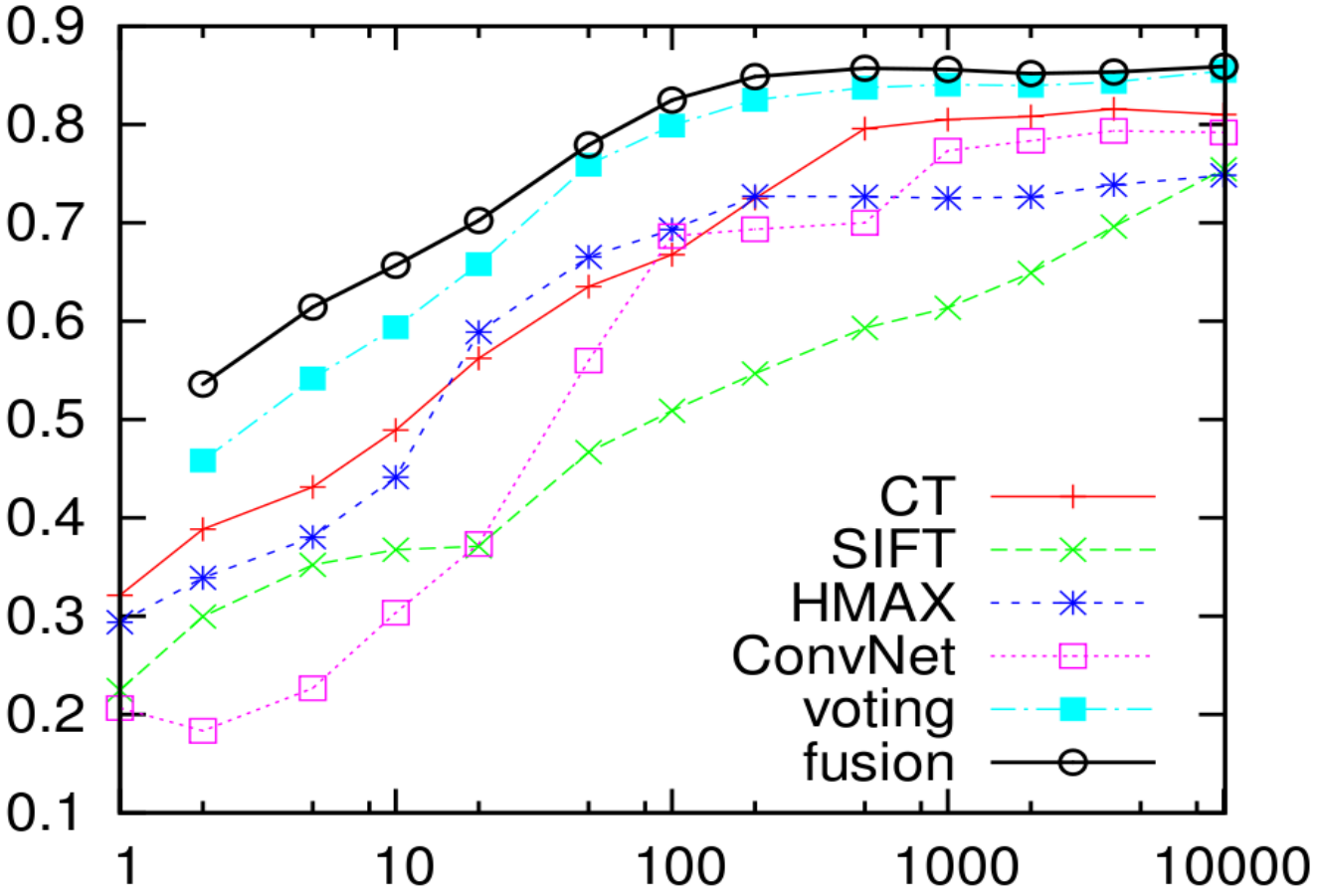
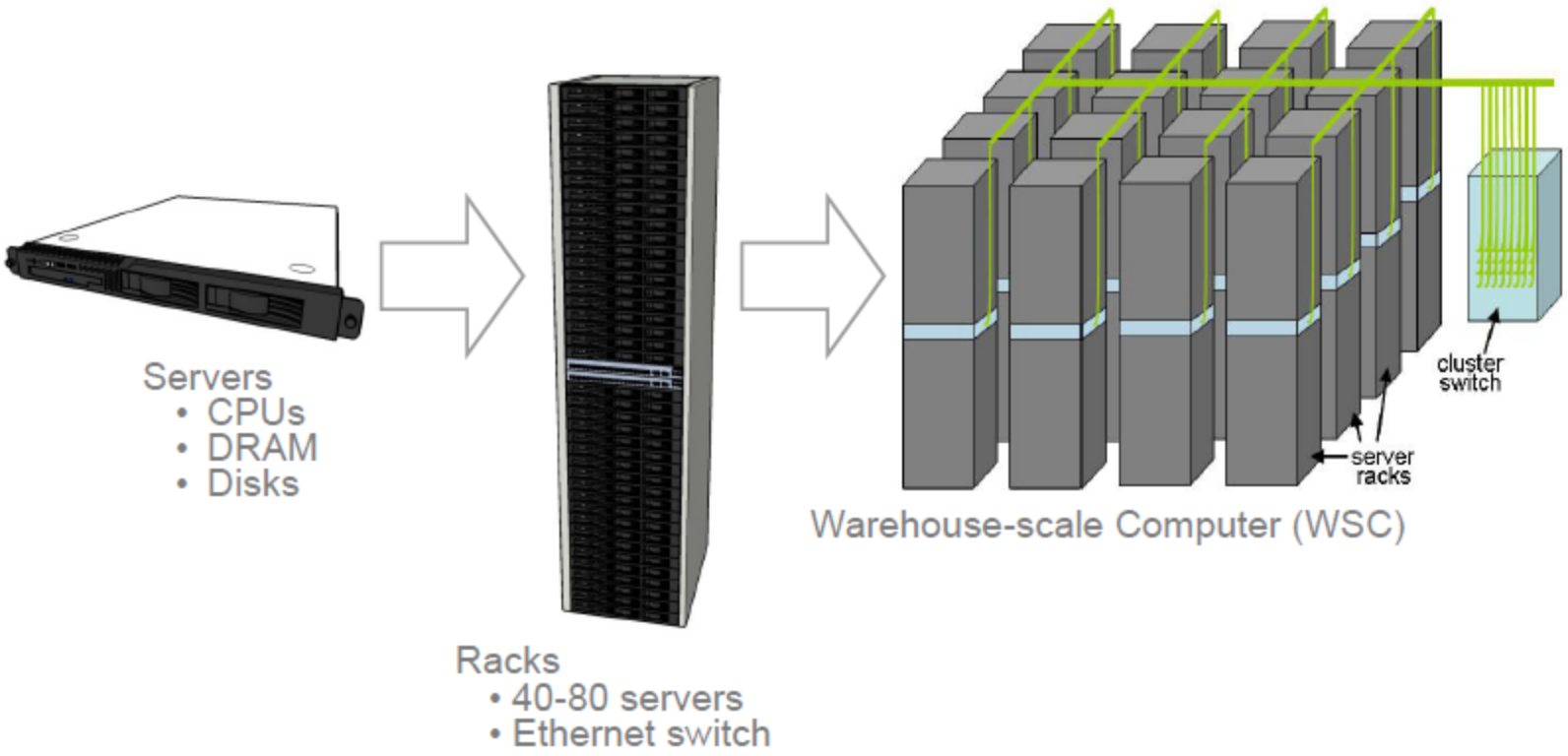


Fig. 14: Accuracy for PicasaWeb dataset. X-axis shows the number of training instances for each category. Y-axis shows the average accuracy.

Data Center (Cloud)



Sample Hierarchy

- Server
  - 16GB DRAM; 160 GB SSD; 5 x 1TB disk
- Rack
  - 40 servers
  - 48 port Gigabit Ethernet switch
- Warehouse
  - 10,000 servers (250 racks)
  - 2K port Gigabit Ethernet switch

Syllabus

- Feature Extraction (Lecture #2)
- Machine Learning (Lecture #3)
  - Small sample pool
  - Large sample pool
- Similarity (Lecture #4)
- Multimodal Fusion (Lecture #5)
- Indexing (Lecture #6)
- Scalability (Lectures #7 & #8)
  - Parallel algorithms
  - Online algorithms

Reading

- Foundations of Large-Scale Multimedia Information Management and Retrieval, E. Y. Chang, Springer, 2011
  - Chapter #1